

Tutorial No.12

Q.1 Solve the following:

1. An inductor of 0.5H is connected in series with a resistance of 6Ω and a capacitor of 0.02 farad and a voltage generator having alternating voltage $24\sin 10t$. Find the charge and current at any time t if at $t=0$, $Q=I=0$.
2. A circuit consists of an inductance of 0.05 henry, a resistance of 20Ω and a condenser of capacity 100 microfarad and an emf of $E = 100$ V. At $t=0$, $Q=I=0$. Find Q and I at any time t .
3. A circuit consists of an inductance of 0.1 henry, a resistance of 20Ω and a condenser of capacity 100 microfarads. At $t=0$, $Q=0.05$ and $I=0$. Find Q and I at any time t .
4. A circuit consists of an inductance of 0.1 henry, a resistance of 20Ω and a condenser of capacity 25 microfarads. At $t=0$, $Q=0.05$ and $I=0$. Find Q and I at any time t .
5. An inductor of 0.25H is connected in series with negligible resistance and a capacitor of 0.04 farad and a voltage generator having alternating voltage $12\sin 10t$. Find the charge and current at any time t if at $t=0$, $Q=I=0$.

Q.2 Find the solution to the following system:

1. $x'_1 = x_1 + 2x_2$ and $x'_2 = 3x_1 + 2x_2$ where $x_1(0) = 0$, $x_2(0) = -4$
2. $x'_1 = 2x_1 + 1x_2$ and $x'_2 = 1x_1 + 2x_2$ where $x_1(0) = 1$, $x_2(0) = 2$
3. $y' = Ay = \begin{bmatrix} -3 & 1 \\ 1 & -3 \end{bmatrix} y$
4. $y' = Ay = \begin{bmatrix} 3 & 1 \\ 5 & 1 \end{bmatrix} y$
5. $y' = Ay = \begin{bmatrix} 3 & 9 \\ 4 & -3 \end{bmatrix} y$
6. $y' = Ay = \begin{bmatrix} 1 & 2 \\ 3 & 2 \end{bmatrix} y$